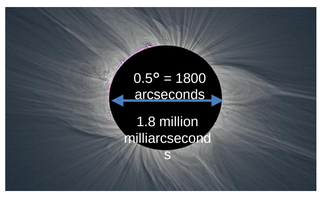
Toward Attaining the Diffraction Limit and Beyond in Astronomical Imaging

Farzad Kamalabadi, Ulas Kamaci, Zifei Han

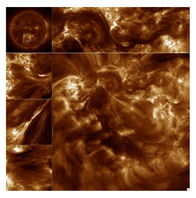
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INTRODUCTION

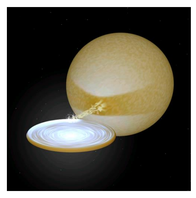
Why is milliarcsecond (mas) resolution coveted? Many astrophysical phenomena as seen from Earth are known from observation or inferred from theory to subtend ~1-100 mas.



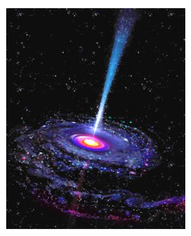
1 mas = 0.000000278 degree
= 0.725 km at the distance of the Sun
= 8 au at the distance of the galactic center



X-ray binary system (Credit: Vdsluys)

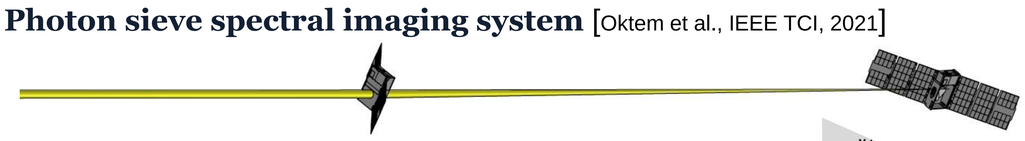


Magnetic structure in the Sun's corona (Credit: A. Winebarger and the Hi-C Team)

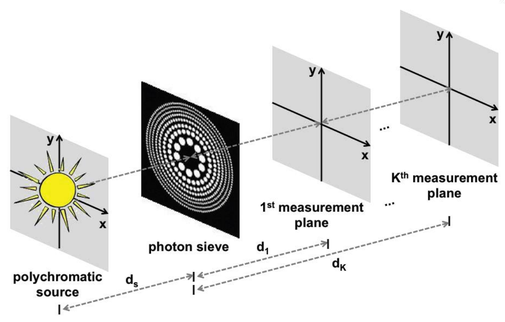


Jet from a supermassive black hole (Credit: NASA)

Photon sieve spectral imaging system [Oktem et al., IEEE TCI, 2021]

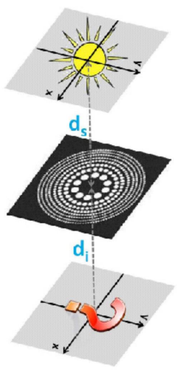


- K measurement planes
- A polychromatic source of P spatially and mutually incoherent monochromatic sources



MODELS

Image formation model



$\mathcal{F}$: Free space (Fresnel propagation)

$$u(x,y) = s(x,y) * \frac{1}{i\lambda d_s} e^{i\pi \frac{x^2+y^2}{\lambda d_s}}$$

$\mathcal{P}$: Passage through photon sieve

$$v(x,y) = \sum_n \underbrace{a_n(x,y)}_{\text{aperture func. of } n\text{th pinhole}} u(x,y)$$

$\mathcal{F}$: Free space (Fresnel propagation)

$$i(x,y) = v(x,y) * \frac{1}{i\lambda d_i} e^{i\pi \frac{x^2+y^2}{\lambda d_i}}$$

[Oktem, Davila, Kamalabadi, *IEEE ICIP* 2014]

$$i(x,y) = -\frac{d_s}{d_i} e^{i\pi \frac{x^2+y^2}{\lambda d_i}} \left[\tilde{s}(x,y) * g_{\lambda,d_i}(x,y)\right]$$

- $\tilde{s}(x,y) = s\left(-\frac{d_s}{d_i}x, -\frac{d_s}{d_i}y\right) e^{\frac{i\pi d_s (x^2+y^2)}{\lambda d_i^2}}$
- $g_{\lambda,d_i}(x,y) = i\frac{\lambda}{\Delta} e^{-i\pi \frac{x^2+y^2}{\Delta \lambda d_i^2}} * \sum_n A_n\left(\frac{x}{\lambda d_i}, \frac{y}{\lambda d_i}\right)$
- $A_n(f_x, f_y)$: Fourier Transform of $a_n(x,y)$ (i.e. jinc function)
- $\Delta = 1/d_i + 1/d_s$

Forward model

- It relates the intensities of multiple monochromatic sources to the superimposed images formed at the measurement planes.

Image formation model for multiple sources

$$t_k(x,y) = \sum_{p=1}^{P} \bar{s}_p(x,y) * \bar{g}_{\lambda_p,d_k}(x,y)$$

$t_k(x,y)$

- intensity observed at the kth measurement plane

$$\bar{s}_p(x,y) = \frac{d_s^2}{d_k^2} f_{\lambda_p}\left(-\frac{d_s}{d_k}x, -\frac{d_s}{d_k}y\right)$$

- scaled intensity of the pth source

$$\bar{g}_{\lambda_p,d_k}(x,y) = \left| i\frac{\lambda_p}{\Delta_k} e^{-i\pi \frac{x^2+y^2}{\Delta_k \lambda_p d_k^2}} * A\left(\frac{x}{\lambda_p d_k}, \frac{y}{\lambda_p d_k}\right) \right|^2$$

- incoherent point spread function (PSF) of the photon sieve at wavelength λ_p and distance d_k ($A = \sum_n A_n$)

Discrete forward model for multiple sources

- Bandlimited (diffraction-limited) versions of the input functions ($\tilde{s}_p$) will be recovered for a bandlimited PSF.
- After discretization with sinc basis:

$$t_k[m,n] = \sum_{p=1}^{P} \tilde{s}_p[m,n] * \bar{g}_{\lambda_p,d_k}[m,n], \quad m,n = 0,\dots,N-1$$

or in matrix-vector form

$$t_k = \sum_{p=1}^{P} H_{k,p}\tilde{s}_p$$

- With measurements from all planes and white Gaussian noise:

$$\mathbf{y} = \mathbf{H}\tilde{\mathbf{s}} + \mathbf{n}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_{1,1} & \dots & \mathbf{H}_{1,P} \\ \vdots & & \vdots \\ \mathbf{H}_{K,1} & \dots & \mathbf{H}_{K,P} \end{bmatrix}$$

MAP Fomulation

MAP estimate of $\tilde{\mathbf{s}}$ from measurements $\mathbf{y}$

$$\arg\max_{\tilde{\mathbf{s}}} p(\mathbf{y} \mid \tilde{\mathbf{s}})\, p(\tilde{\mathbf{s}})$$

- $p(\mathbf{y}|\tilde{\mathbf{s}})$: conditional pdf of $\mathbf{y}$ given $\tilde{\mathbf{s}}$
- $p(\tilde{\mathbf{s}})$: prior distribution of $\tilde{\mathbf{s}}$

Equivalent regularized least square problem:

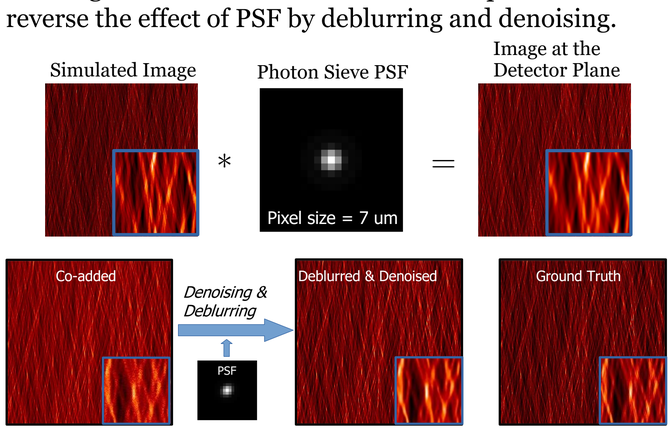
$$\arg\max_{\tilde{\mathbf{s}}} ||\mathbf{y} - \mathbf{H}\tilde{\mathbf{s}}||_W^2 + \alpha^2 R(\tilde{\mathbf{s}})$$

with $\quad R(\tilde{\mathbf{s}}) \propto -\log p(\tilde{\mathbf{s}})$

EXAMPLE

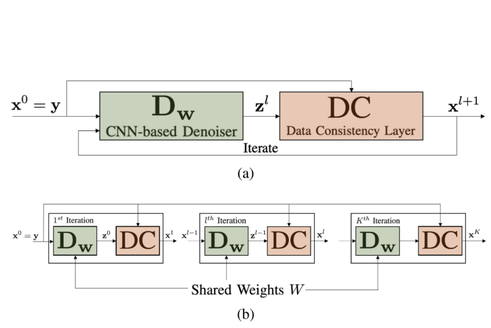
Image Reconstruction for a Monochromatic Source

- First row: the scene gets convolved with the PSF of the photon sieve.
- Second row: reverse the effect of PSF by deblurring and denoising.

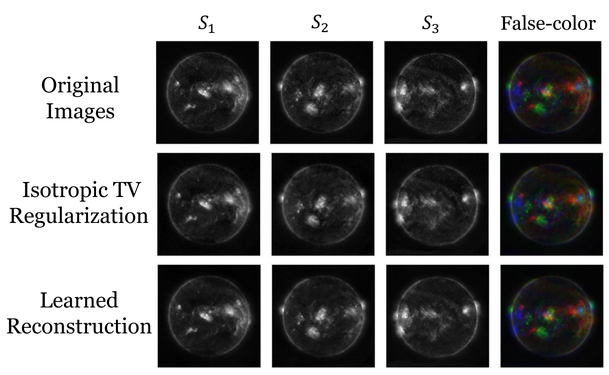


LEARNED RECONSTRUCTION WITH DEEP PRIORS

Learn a 2D prior through a deep network that is end-to-end trained through unrolling, with half-quadratic splitting applied, to simultaneously incorporate physics-based forward model and the data. [Oktem et al., 2021]



Model-based learned reconstruction framework



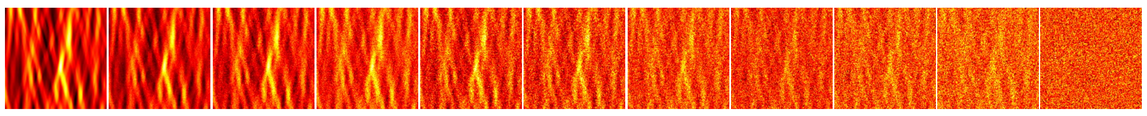
Reconstruction results for three spectral sources, with additional false-color representations.

Score-Based Prior

Diffusion models define the generative process as the *reverse* of the noising process. Define the stochastic differential equation (SDE) for the data noising process x(t), t ∈ [o, T].

Forward SDE (data→noise)

$$\mathbf{x}(0) \longrightarrow d\boldsymbol{x} = -\frac{\beta(t)}{2}\boldsymbol{x}\,dt + \sqrt{\beta(t)}\,d\boldsymbol{w} \longrightarrow \mathbf{x}(T)$$



$$\mathbf{x}(0) \longleftarrow d\boldsymbol{x} = \left(-\frac{\beta(t)}{2}\boldsymbol{x} - \beta(t)\nabla_{\boldsymbol{x}} \log p_t(\boldsymbol{x}_t)\right) dt + \sqrt{\beta(t)}\,d\bar{\boldsymbol{w}} \longleftarrow \mathbf{x}(T)$$

Reverse SDE (noise→data)

Train a neural network s_θ to approximate the score function $\nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t)$:

$$\theta^* = \arg\min_\theta \mathbb{E}_{t\sim\mathcal{U}(\epsilon,1),\, \boldsymbol{x}(t)\sim p(\boldsymbol{x}(t)|\boldsymbol{x}(0)),\, \boldsymbol{x}(0)\sim p_{\text{data}}} \left\| s_\theta(\boldsymbol{x}(t), t) - \nabla_{\boldsymbol{x}_t} \log p(\boldsymbol{x}(t) \mid \boldsymbol{x}(0)) \right\|_2^2$$

Posterior Sampling

Once the prior is trained, modify reverse SDE to incorporate y:

$$\nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t) \xrightarrow[\text{on measurements}]{\text{Condition}} \nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t \mid \boldsymbol{y}) = \nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t) + \nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{y} \mid \boldsymbol{x}_t)$$

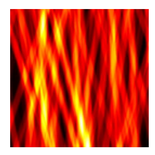
$$p(\boldsymbol{y} \mid \boldsymbol{x}_t) \approx p(\boldsymbol{y} \mid \hat{\boldsymbol{x}}_0), \quad \text{where} \quad \hat{\boldsymbol{x}}_0 \simeq \left(\boldsymbol{x}_t + (1-\bar{\alpha}(t))\, \boldsymbol{s}_{\theta^*}(\boldsymbol{x}_t, t)\right) / \sqrt{\bar{\alpha}(t)}$$

$$\nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t \mid \boldsymbol{y}) \simeq s_{\theta^*}(\boldsymbol{x}_t, t) - \rho \nabla_{\boldsymbol{x}_t} \left\| \boldsymbol{y} - \mathcal{A}(\hat{\boldsymbol{x}}_0) \right\|_2^2$$

Additional gradient descent step during sampling ensures data consistency.

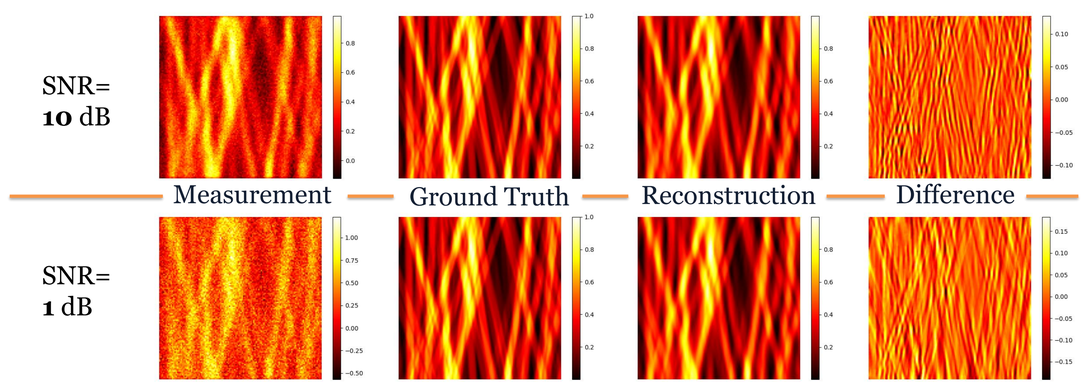
Dataset

The training dataset consists of 10000 nanoflare synthetic images of randomly oriented, vertical-like strands, with the number of strands, the strand thickness, and the maximum bend angle specified on the right.



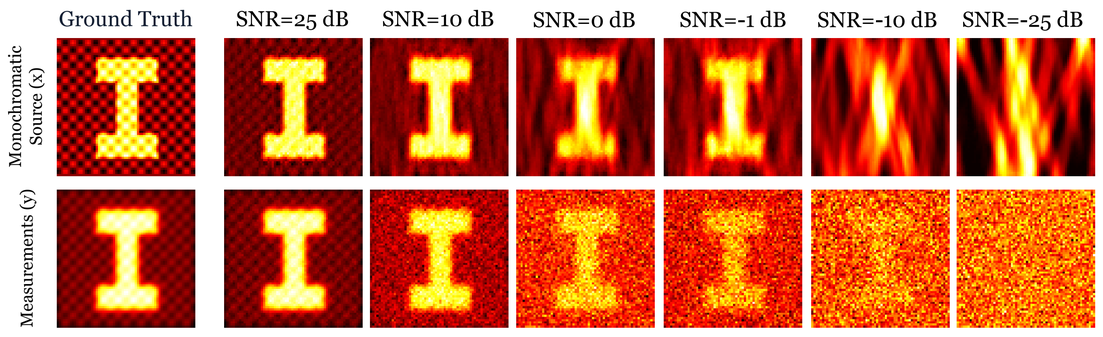
Num of Strands = 100
Thickness = 22 pixels
Max Bend Angle = 20°

RESULTS



The reconstruction performance for a test image with two different noise levels. The last column shows the difference between ground truth and the reconstruction results.

Out-of-Distribution Reconstruction Comparison



The reconstruction performance for an out-of-distribution image with several different noise levels. As the noise level increases, the reconstruction results look more like the prior.